Formula and Constants Sheet for EEME30004 Communications and Networks

$$\beta = \frac{2\pi}{\lambda}$$

Boltzmann constant: $k = 1.38 \times 10^{-23}\, J \cdot K^{-1}$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$\text{RL} = -20\log|\Gamma| \text{ dB}$$

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$$

$$\lambda = \frac{c}{f\sqrt{\varepsilon_{eff}}}$$

$$T_e = \frac{N_o}{kB}$$

$$F = \frac{SNR_i}{SNR_o} = \frac{S_i/N_i}{S_o/N_o} = \frac{N_o}{A_p \times N_i}$$

$$N_i = kT_0B$$

$$T_e = (F-1)T_0$$

$$N_o = kGB(T_0 + T_e)$$

$$F_{\text{cas}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1G_2} + \cdots$$

$$F = 1 + (L-1)\frac{T}{T_0} \text{ (L is loss)}$$

Get familiar with the conversion from dB to linear, from dBm to mW

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|}$$

$$|\Delta| = |S_{11}S_{22} - S_{12}S_{21}|$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12}S_{21}|}$$

$$IIP3_{\text{cas}} = \frac{1}{\frac{1}{IIP3_1} + \frac{G_1}{IIP3_2} + \frac{G_1G_2}{IIP3_3} + \frac{G_1G_2G_3}{IIP3_4} + \frac{G_1G_2G_3G_4}{IIP3_5} + \cdots} \quad \text{(mW)}$$